

Listing 14.2: Minitab output for intrablock analysis of lithium data.

Source	DF	Seq SS	Adj SS	Seq MS	F	P
week	1	0.031974	0.031974	0.031974	15.79	0.004
subject	11	0.039344	0.029946	0.003577	1.77	0.215
treatmen	3	0.005603	0.005603	0.001868	0.92	0.473
Error	8	0.016203	0.016203	0.002025		

mean square for the week blocking factor is fairly large. If we had ignored the week effect, we could anticipate an error mean square of

$$\frac{11 \times .0020253 + .031974}{12} = .00452 ,$$

more than doubling the error mean square in the Youden Square design.

14.3 Partially Balanced Incomplete Blocks

BIBD's are too big
for some g and k

BIBD's are great, but their balancing requirements may imply that the smallest possible BIBD for a given g and k is too big to be practical. For example, let's look for a BIBD for $g = 12$ treatments in incomplete blocks of size $k = 7$. To be a BIBD, $\lambda = r(k - 1)/(g - 1) = 6r/11$ must be a whole number; this implies that r is some multiple of 11. In addition, $b = rg/k = (11 \times m) \times 12/7$ must be a whole number, and that implies that b is a multiple of $11 \times 12 = 132$. So the smallest possible BIBD has $r = 77$, $b = 132$, and $N = 924$. This is a bigger experiment that we are likely to run.

PBIBD has
 $N = gr = bk$;
some treatment
pairs more
frequent

Partially Balanced Incomplete Block Designs (PBIBD) allow us to run incomplete block designs with fewer blocks than may be required for a BIBD. The PBIBD has g treatments and b blocks of k units each; each treatment is used r times, and there is a total of $N = gr = bk$ units. The PBIBD does not have the requirement that each pair of treatments occurs together in the same number of blocks. This in turn implies that not all differences $\hat{\alpha}_i - \hat{\alpha}_j$ have the same variance in a PBIBD.

Sample PBIBD

Here is a sample PBIBD with $g = 12$, $k = 7$, $r = 7$, and $b = 12$. In this representation, each row is a block, and the numbers in the row indicate which treatments occur in that block.

Block	Treatments						
1	1	2	3	4	5	8	10
2	2	3	4	5	6	9	11
3	3	4	5	6	7	10	12
4	1	4	5	6	7	8	11
5	2	5	6	7	8	9	12
6	1	3	6	7	8	9	10
7	2	4	7	8	9	10	11
8	3	5	8	9	10	11	12
9	1	4	6	9	10	11	12
10	1	2	5	7	10	11	12
11	1	2	3	6	8	11	12
12	1	2	3	4	7	9	12

We see, for example, that treatment 1 occurs three times with treatments 5 and 9, and four times with all other treatments.

The design rules for a PBIBD are fairly complicated:

1. There are g treatments, each used r times. There are b blocks of size $k < g$. Of course, $bk = gr$. No treatment occurs more than once in a block.
2. There are m *associate classes*. Any pair of treatments that are i th associates appears together in λ_i blocks. We usually arrange the λ_i values in decreasing order, so that first associates appear together most frequently.
3. All treatments have the same number of i th associates, namely ρ_i .
4. Let A and B be two treatments that are i th associates, and let p_{jk}^i be the number of treatments that are j th associates of A and k th associates of B. This number p_{jk}^i does not depend on the pair of i th associates chosen. In particular, $p_{jk}^i = p_{kj}^i$.

Requirements for
PBIBD

Associate classes

ρ_i i th associates

The PBIBD is partially balanced, because the variance of $\hat{\alpha}_i - \hat{\alpha}_j$ depends upon whether i, j are first, second, or m th associates. The randomization of a PBIBD is just like that for a BIBD.

Randomize
PBIBD like BIBD

Let's check the design given above and verify that it is a PBIBD. First note that $g = 12$, $k = 7$, $r = 7$, $b = 12$, and no treatment appears twice in a block. Next, there are two associate classes, with first associates appearing together four times and second associates appearing together three times. The pairs (1,5), (1,9), (2,6), (2,10), (3,7), (3,11), (4,8), (4,12), (5,9), (6,10), (7,11), and (8,12) are second associates; all other pairs are first associates. Each treatment has nine first associates and two second associates. For any pair of

first associates, there are six other treatments that are first associates of both, four other treatments that are first associates of one and second associates of the other (two each way), and no treatments that are second associates of both. We thus have

$$\{p_{ij}^1\} = \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}.$$

For any pair of second associates, there are nine treatments that are first associates of both, and one treatment that is a second associate of both, so that

$$\{p_{ij}^2\} = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus all the design requirements are met, and the example design is a PBIBD.

Intrablock analysis is treatments adjusted for blocks

One historical advantage of the PBIBD was that the analysis could be done by hand. That is, there are relatively simple expressions for the various intra- and interblock analyses. With computers, that particular advantage is no longer very useful. The intrablock analysis of the PBIBD is simply treatments adjusted for blocks, as with the BIBD.

PBIBD less efficient on average than BIBD

The efficiency of a PBIBD is actually an average efficiency. The variance of $\hat{\alpha}_i - \hat{\alpha}_j$ depends on whether treatments i and j are first associates, second associates, or whatever. So to compute efficiency $E_{\text{PBIBD:RCB}}$, we divide the variance obtained in an RCB for a pairwise difference ($2\sigma^2/r$) by the average of the variances of all pairwise differences in the PBIBD. There is an algorithm to determine $E_{\text{PBIBD:RCB}}$, but there is no simple formula. We can say that the efficiency will be less than $g(k-1)/[(g-1)k]$, which is the efficiency of a BIBD with the same block size and number of treatments.

There are several extensive catalogues of PBIBD's, including Bose, Clatworthy, and Shrikhande (1954) (376 separate designs) and Clatworthy (1973).

14.4 Cyclic Designs

Cyclic designs are simple

Cyclic designs are easily constructed incomplete block designs that permit the study of g treatments in blocks of size k . We will only examine the simplest situation, where the replication r for each treatment is a multiple of k , the block size. So $r = mk$, and $b = mg$ is the number of blocks. Cyclic designs include some BIBD and PBIBD designs.

Cycles of treatments

A cycle of treatments starts with an initial treatment and then proceeds through the subsequent treatments in order. Once we get to treatment g , we go back down to treatment 1 and start increasing again. For example, with seven treatments we might have the cycle (4, 5, 6, 7, 1, 2, 3).